

CA - Prerequisites Document

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TITLE: Comic Collection and Review Platform

INTRODUCTION

The Comic Collection and Review Platform is a full-stack web application developed to help users manage their comic collections, write reviews, and view previously added comics. The frontend is built using Angular and TypeScript, while the backend is powered by Flask, and data is stored in MongoDB. This document outlines the necessary system requirements, technology stack, and setup instructions to run the project successfully.

SYSTEM REQUIREMENTS

1. Hardware Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 8GB (Recommended: 16GB for optimal performance)
- Storage: At least 10GB of free disk space
- Operating System: Windows 10/11, macOS, or Linux

2. Software Requirements

- Node.js (v18.16.1 or later) – For running the Angular frontend
 - npm (v9.5.1 or later) – For managing frontend dependencies
 - Python (3.11.9 or later) – For running the Flask backend
 - MongoDB Atlas – Cloud-based NoSQL database
 - Visual Studio Code – Recommended IDE for both frontend and backend development
 - Postman – For testing API endpoints
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TECHNOLOGY STACK

1. Frontend

- Angular – Component-based framework for building UIs
- TypeScript – Strongly typed JavaScript for scalable code
- CSS – For custom styling
- Flask – Used to bridge communication between frontend and backend

2. Backend

- Flask – Lightweight Python web framework for handling API requests
- Flask-CORS – To manage cross-origin requests
- PyMongo – Python library to connect Flask with MongoDB

3. Database

- MongoDB Atlas – A cloud-based NoSQL database to store comic and review data

PROJECT STRUCTURE

```
▼ COMIC-COLLECTION-MAIN
  ▼ Back-end code
    ▼ __pycache__
      app.cpython-313.pyc
    app.py
  ▼ Front-end code
    > .angular
    > .vscode
    > node_modules
  ▼ src
    ▼ app
      # addcomic.component.css
      <> addcomic.component.html
      TS addcomic.component.ts
      # app.component.css
      <> app.component.html
      TS app.component.spec.ts
      TS app.component.ts
      TS app.config.ts
      TS app.module.ts
      TS app.routes.ts
      # collections.component.css
      <> collections.component.html
      TS collections.component.ts
      # comic.component.css
      <> comic.component.html
      TS comic.component.ts
      # comics.component.css
      <> comics.component.html
      TS comics.component.ts
      # editcomic.component.css
      <> editcomic.component.html
      TS editcomic.component.ts
      # editreview.component.css
      <> editreview.component.html
      TS editreview.component.ts
```

```
▼ COMIC-COLLECTION-MAIN
  ▼ Front-end code
    ▼ src
      ▼ app
        TS search.component.ts
        TS web.service.ts
      ▼ assets
        .gitkeep
        Marvel_comic_collections.png
      > environments
      debug.log
      ★ favicon.ico
      <> index.html
      TS main.ts
      TS polyfills.ts
      # styles.css
      TS test.ts
      .editorconfig
      .gitignore
      {} angular.json
      {} package-lock.json
      {} package.json
      ⓘ README.md
      {} tsconfig.app.json
      tsconfig.json
      {} tsconfig.spec.json
    ▼ venv
      > Include
      > Lib
      > Scripts
      .gitignore
      pyvenv.cfg
      {} package-lock.json
      JS playground-1.mongoose.js
      ⓘ README.md
```

CONCLUSION:

The **Comic Collection and Review Platform** provides a seamless solution for comic enthusiasts to manage, review, and revisit their favorite comic books. By integrating **Angular** with **TypeScript** on the frontend and using **Flask** with **MongoDB** on the backend, the project showcases the power of modern full-stack development. The use of cloud-based MongoDB Atlas ensures scalability and efficient data storage. With a user-friendly interface and robust backend logic, this platform offers a smooth and interactive experience for users, highlighting the effectiveness of combining powerful technologies in a cohesive system.